

System Architecture — Final Design Correction v2

Authoritative status: This page supersedes any earlier architecture wording that conflicts with the approved requirements, ERD, relational schema, use cases, or decision log. The retained original architecture remains included for traceability.

Area	Final phase-one decision
AI model	obadx/muaalem-model-v3_2 , based on facebook/w2v-bert-2.0, MIT license. Used for Quranic phoneme/articulation analysis. Published model results are evidence, not a guarantee of Itqan accuracy.
Analysis boundary	Recitation analysis runs in a separate Python service on the server. Audio is held in memory for analysis and is discarded; it is not stored in PostgreSQL or exposed to teachers.
Teacher enrollment	A Quran Reader joins a teacher-created LEARNING_CLASS using a join code. There is no invitation-link or acceptance workflow in the phase-one design.
Teacher access	Teacher reads are role-based and restricted to active class enrollment. Teacher-facing views expose stored results such as activity, completion, mastery, and permitted messages—not recitation audio.
Requirements coverage	Architecture traceability covers FR-01 through FR-26 . FR-23 maps to dashboard/progress data; FR-24 to class/enrollment and join-code behavior; FR-25 to teacher messaging; FR-26 to pause events, session handling, and smart prompting.
Evaluation wording	NFR-01 evaluation uses an IS499 evaluation/test-data activity , not formal user research. No survey/interview findings are assumed.
Quran content	Juz Amma content is bundled locally from the selected King Fahd Glorious Quran Printing Complex source: 37 surahs, 564 verses, 2,308 words. Uthmani is used for display and the matching form is used for comparison.
Authentication	Authentication is delegated to a managed identity provider. USER stores the provider-issued identity_subject; no local password hash is stored. The exact vendor remains an implementation detail.
Consent/privacy	CONSENT_RECORD stores per-purpose consent history. Recitation audio is not retained. External processing is not a production deployment assumption and requires vendor/privacy/PDPL transfer review before use.
Hosting	Production hosting/deployment remains deferred to IS499. The architecture defines the component and data boundaries without asserting an unapproved production vendor or location.

Authoritative requirement-to-architecture additions

- FR-23 — Teacher Dashboard: Class Service + Progress Service + stored session/progress/streak data; access restricted by active enrollment.
- FR-24 — Manage Class: Class Service + LEARNING_CLASS/CLASS_ENROLLMENT; learner joins through teacher-provided join code.
- FR-25 — Send Student Messages: Class Service + TEACHER_MESSAGE; access is scoped by class enrollment.
- FR-26 — Receive Smart Prompting: Session Service/Recitation UI + PAUSE_EVENT; triggered by pause, hesitation, or apparent forgetting during recitation.

Source record

Model evidence: https://huggingface.co/obadx/muaalem-model-v3_2 and <https://arxiv.org/abs/2509.00094>. Project decisions are recorded in DECISION_LOG and OPEN_QUESTIONS. This correction does not expand current scope beyond the approved requirements.